

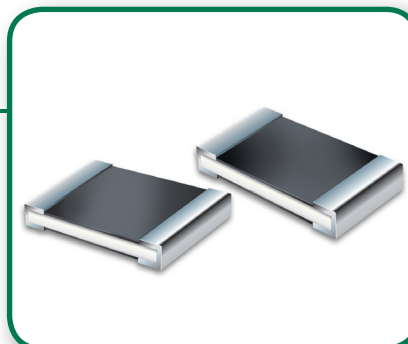
RCT SERIES

CHIP RESISTORS

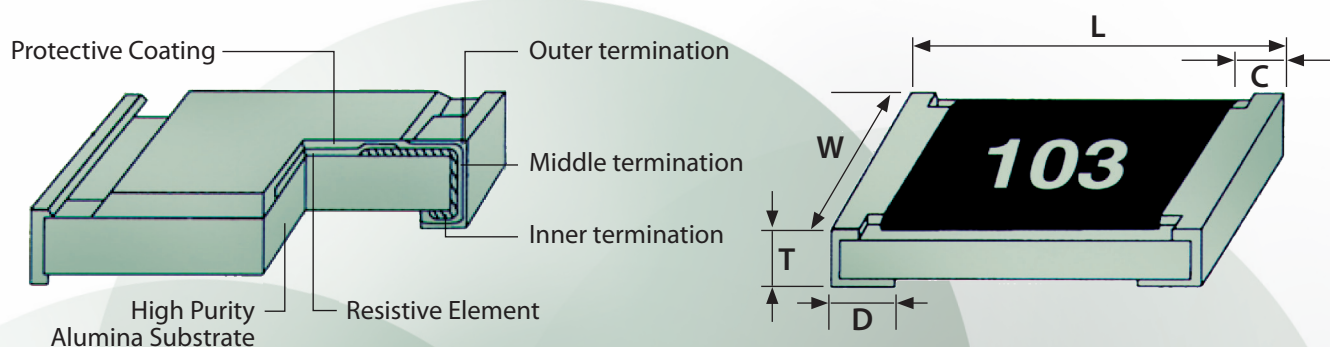
RoHS

FEATURES

- Advance thick film technology
- Wide size from 0201 to 2512
- Wide resistance range
- Compatible with wave and reflow soldering



CONFIGURATION



DIMENSIONS

SIZE	DIMENSIONS (mm)				
	L	W	T	C	D
0201	0.60 ±0.03	0.30 ±0.03	0.23 ±0.03	0.15 ±0.05	0.15 ±0.05
0402	1.00 ±0.10	0.50 ±0.05	0.35 ±0.05	0.20 ±0.10	0.25 ±0.10
0603	1.60 ±0.10	0.80 ±0.10	0.45 ±0.10	0.25 ±0.15	0.30 ±0.15
0805	2.00 ±0.10	1.25 ±0.10	0.50 ±0.10	0.35 ±0.20	0.30 ±0.20
1206	3.10 ±0.10	1.60 ±0.10	0.55 ±0.10	0.45 ±0.20	0.40 ±0.20
1210	3.10 ±0.10	2.50 ±0.10	0.55 ±0.10	0.45 ±0.25	0.45 ±0.25
1218	3.10 ±0.10	4.60 ±0.10	0.55 ±0.10	0.45 ±0.25	0.45 ±0.25
1812	4.50 ±0.10	3.10 ±0.20	0.55 ±0.05	0.55 ±0.20	0.70 ±0.20
2010	5.00 ±0.20	2.50 ±0.15	0.55 ±0.10	0.60 ±0.25	0.50 ±0.25
2512	6.35 ±0.20	3.10 ±0.15	0.55 ±0.10	0.60 ±0.25	0.60 ±0.25



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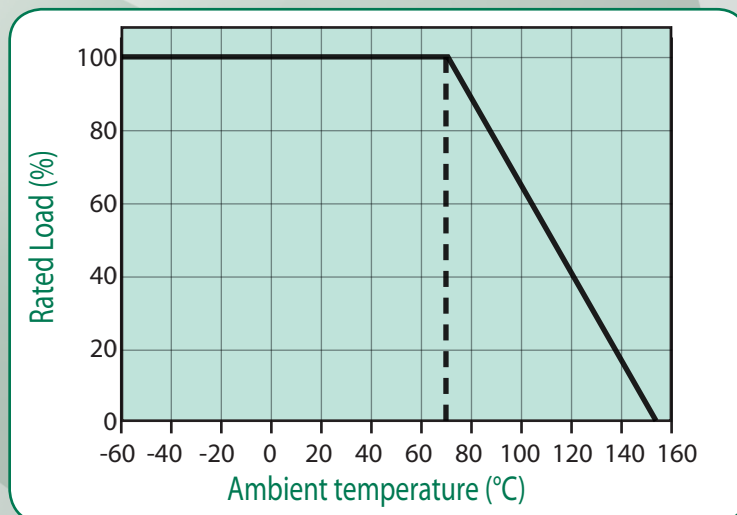
RoHS

RATING RCT SERIES

Type	RCT01	RCT02	RCT03	RCT05	RCT06	RCT10	RCT18	RCT12	RCT20	RCT25
Size	0201	0402	0603	0805	1206	1210	1812	1218	2010	2512
Power Rating at 70°C	1/20W	1/16W	1/10W	1/8W	1/4W	1/3W	1/2W	1W	1/2W	1W
Max RCWV	15V	50V	50V	150V	200V	200V	200V	200V	200V	200V
Max Overload Voltage	30V	100V	100V	300V	400V	400V	400V	400V	400V	400V
Jumper<50mΩ	0.5A	1A	1A	1A	2A	2A	2A	2A	2A	2A
Operating Temp. Range	-55~+155°C									
Standard	0Ω	0~50mΩ								
Resistance	±1% (E-96)	1Ω~2.2MΩ		1Ω~2.2MΩ						
Range	±5% (E-24)	1Ω~3.3MΩ		1Ω~22MΩ						

The working voltage is calculated based on the resistance value following the formula of $V=\sqrt{(P \cdot R)}$ or to its maximum extent as indicated above.
The overload voltage is calculated based on the resistance value following the formula of $V=2.5\sqrt{(P \cdot R)}$ or to its maximum extent as indicated above.

POWER DERATING CURVE



For resistors operated in ambient temperatures above 70°C, power rating must be derated in accordance with the curve below.



RCT SERIES

CHIP RESISTORS

RoHS

SPECIFICATION AND TEST METHODS

ITEM	SPECIFICATION	TEST METHOD
Solderability	Over 95% of termination must be covered with solder	JIS C 5201 4.17 / MIL-STD-202G METHOD 208H / IEC 60115-1 4.17 Reflow Soldering: Bath temperature: (235±5)°C, Immersion time: (2±0.5)s WAVE Soldering: Bath temperature: (270±10)°C, Immersion time: (2±0.5)s
Resistance to Solder Heat	J - G: $\Delta R \leq \pm(1\% + 0.1\Omega)$ F: $\Delta R \leq \pm(0.5\% + 0.05\Omega)$ No mechanical damage	JIS C 5201 4.18 / MIL-STD-202G METHOD 210F / IEC 60115-1 4.18 The temperature of the solder bath shall be (260±5)°C, The immersion time shall be (5±0.5)s or (10±1)s as prescribed by the detail specification
Temperature Coefficient of Resistance (TCR)	0402 (0402 & 0201): 1Ω~10Ω; >3.3M: ±400ppm/°C 10Ω(excluding 10Ω)~3.3M: ±200ppm/°C 0603 (0603 or above): 1Ω~10Ω; >1M: ±200ppm/°C 10Ω(excluding 10Ω)~1M: ±100ppm/°C	JIS C 5201 4.8 / MIL-STD-202G METHOD 304 / IEC 60115-1 4.8 The temperature characteristic or temperature coefficient of resistance is not specified for resistance values of less than 5R owing to difficulty of accurate measurement
Short time Overload	J - G: $\Delta R \leq \pm(2\% + 0.1)$ F: $\Delta R \leq \pm(1\% + 0.05\Omega)$	JIS C 5201 4.13/IEC 60115-1 4.13 2.5×Rated voltage or Max. Overload Voltage for 5 sec. measure resistance after 30 minutes
Load Life Humidity	J - G: $\Delta R \leq \pm(3\% + 0.1\Omega)$ F: $\Delta R \leq \pm(1\% + 0.05\Omega)$	JIS C 5201 4.24/MIL-STD-202G Method 106G / IEC 60115-1 4.24 Maintain the temperature of the resistor at 40±2°C and 90-95% RH with the rated voltage applied. Cycle ON for 1.5 hours and OFF for 0.5 hour for 1000+48/-0 hours. After 1-4 hour, measure the resistance value
Load Life	J - G: $\Delta R \leq \pm(3\% + 0.1)$ F: $\Delta R \leq \pm(1\% + 0.05)$	JIS C 5201 4.25/MIL-STD-202G Method 108A / IEC 60115-1 4.25 The resistors shall be subjected to an endurance test of 42 days (1000h) at an ambient temperature of between 15°C and 35°C
Temperature Cycle	J - G: $\Delta R \leq \pm(1\% + 0.1\Omega)$ F: $\Delta R \leq \pm(0.5\% + 0.05)$ No mechanical damage	JIS C 5201 4.19/MIL-STD-202G Method 107G / IEC 60115-1 4.19 Repeat 5 cycles as follows -65°C (30 min.) +25°C (2-3 mi.) +125°C(30 min.) +25°C (2-3 min.)
Insulation Resistance	Between termination and coating must be over 1000MΩ	JIS C 5201 4.6 / MIL-STD-202G Method 302 / IEC 60115-1 4.6 Test voltage: 100±15V
Bending Strength	J - G: $\Delta R \leq \pm(1\% + 0.1\Omega)$ F: $\Delta R \leq \pm(0.5\% + 0.05\Omega)$ No mechanical damage	JIS C 5201 4.33 / MIL-STD-202G Method 211A / IEC 60115-1 4.33 Resistance change after bended on the 90mm PCB. Bend: 3mm for 0201 - 0402 - 0603 - 0805 - 1206 2mm for 1210 - 1812 - 1218 1mm for 2010 - 2512

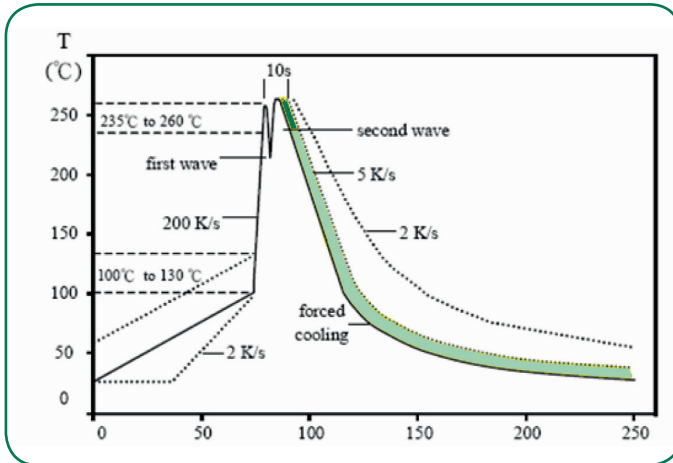


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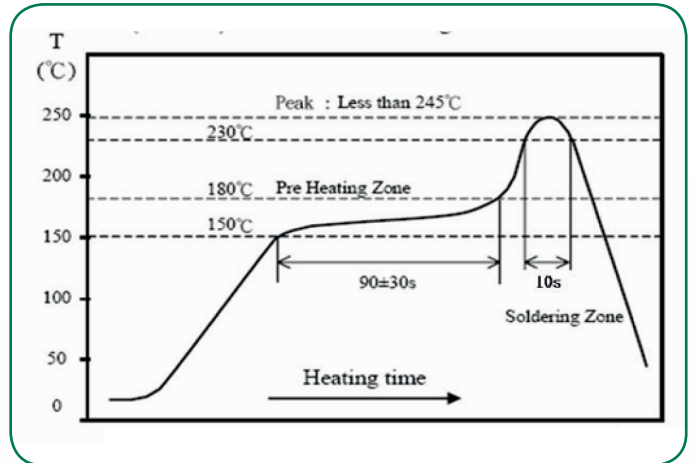
RoHS

SOLDERING TEMPERATURE CURVE



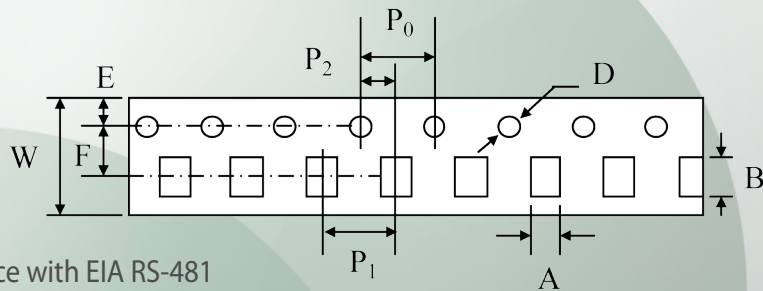
Wave Soldering

Typical values: Solid line Process limits: Dotted line



IR Reflow Soldering

TAPE AND REEL PACKAGE



Taping is in accordance with EIA RS-481

Accumulated dimensional tolerance 40±0.2mm

SIZE	A	B	W	F	E	P ₁	P ₂	P ₀	D
0201	0.40 ±0.10	0.75 ±0.10	8.00 ±0.30	3.50 ±0.05	1.75 ±0.10	2.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
0402	0.65 ±0.10	1.15 ±0.10	8.00 ±0.30	3.50 ±0.05	1.75 ±0.10	2.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
0603	1.10 ±0.20	1.90 ±0.20	8.00 ±0.30	3.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
0805	1.65 ±0.20	2.40 ±0.20	8.00 ±0.30	3.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
1206	2.00 ±0.20	3.50 ±0.20	8.00 ±0.30	3.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
1210	2.85 ±0.20	3.50 ±0.20	12.0 ±0.30	3.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
1218	5.50 ±0.20	3.50 ±0.20	12.0 ±0.30	3.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
1812	3.50 ±0.20	5.50 ±0.20	12.0 ±0.30	3.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
2010	2.80 ±0.20	5.50 ±0.20	12.0 ±0.30	5.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0
2512	3.50 ±0.20	6.70 ±0.20	12.0 ±0.30	5.50 ±0.05	1.75 ±0.10	4.00 ±0.10	2.00 ±0.05	4.00 ±0.10	1.50 +0.10/-0

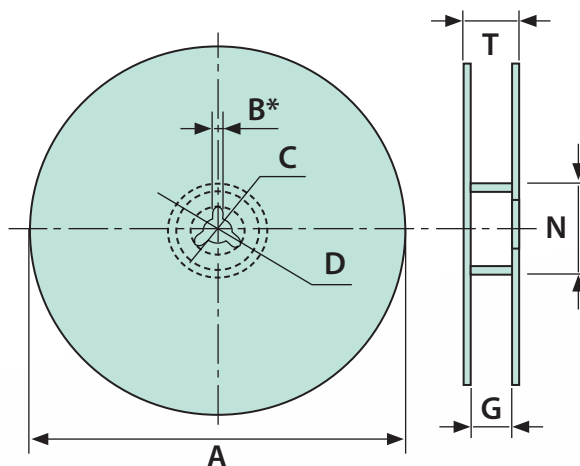


RCT SERIES

CHIP RESISTORS

RoHS

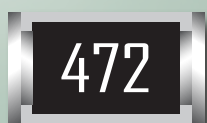
SIZE	PACKAGING QUANTITY
0201	10Kpcs/Reel
0402	10Kpcs/Reel
0603	5Kpcs/Reel
0805	
1206	
1210	
1218	4Kpcs/Reel
1812	
2010	
2512	



SYMBOL	A	N	C	D	B	G	T
Dimension	178 ±2.0	80.0 ±0.5	13.0 ±0.5	20min	2.00 ±0.5	10.0 ±1.5	14.9 max.
	178 ±2.0	80.0 ±0.5	13.0 ±0.5	20min	2.00 ±0.5	13.8 ±1.5	16.7 max.

STORAGE CONDITION		STORAGE TIME
TEMP	HUMIDITY	
10~30°C	30%-75%	1 year

MARKING



±5% (IEC E-24 Series)0603 - 0805 - 1206 (0201 - 0402)

No markings on 0201 and 0402.

Markings on the other sizes are expressed by a 3-digit code in its exact value.

E. G.: 6R8 = 6.8Ω; 100 = 10×10⁰=10Ω; 472 = 47×10²= 4700Ω = 4.7KΩ

±1% (+/-1% Resistance value markings)

a. 0805 - 1206 ±1%

Markings on sizes 0805, 1206 and above are expressed by a 4-digit code.

E. G.: 82R5 = 82.5Ω; 1000=100×10⁰= 100Ω; 2212 = 221×10²= 22100Ω = 22.1KΩ

b. 0603 ±1% (IEC E-96); (10n)

Markings on 0603 ±1% (IEC E-96 Series) are expressed by a 3-digit code;

the first two digits represent the value code and the last capital letter represents the multiplier.

The coding system of the E-96 series is as follows:



RCT SERIES

CHIP RESISTORS

RoHS

100	01	133	13	178	25	237	37	316	49	422	61	562	73	750	85
102	02	137	14	182	26	243	38	324	50	432	62	576	74	768	86
105	03	140	15	187	27	249	39	332	51	442	63	590	75	787	87
107	04	143	16	191	28	255	40	340	52	453	64	604	76	806	88
110	05	147	17	196	29	261	41	348	53	464	65	619	77	825	89
113	06	150	18	200	30	267	42	357	54	475	66	634	78	845	90
115	07	154	19	205	31	274	43	365	55	487	67	649	79	866	91
118	08	158	20	210	32	280	44	374	56	499	68	665	80	887	92
121	09	162	21	215	33	287	45	383	57	511	69	681	81	909	93
124	10	165	22	221	34	294	46	392	58	523	70	698	82	931	94
127	11	169	23	226	35	301	47	402	59	536	71	715	83	953	95
130	12	174	24	232	36	309	48	412	60	549	72	732	84	976	96

(Multiplier code)

CODE	A	B	C	D	E	F	G	X	Y	Z
Multiplier	10 ⁰	10 ¹	10 ²	10 ³	10 ⁴	10 ⁵	10 ⁶	10 ⁻¹	10 ⁻²	10 ⁻³

E. G.: 01Y = 100×10⁻² = 1Ω; 68A = 499×10⁰ = 499Ω; 02D = 102×10³ = 102KΩ

c. 1% E-96, E-24 +/-5%

Any resistance values of 1% tolerance but not included in the E-96 series, the value markings are the same as +/-5% tolerance with labelling identification.

E. G.: 120 = 12×10⁰ = 12Ω; 475 = 47×10⁵ = 4.7MΩ

d. "0", 0~50mΩ

Marking of the 0 Ohm resistor is a '0' with its value range from 0-50mΩ.

No tolerance shown on the product label.

PART NUMBER

RCT	05	10K	J	LF
Type	Size	Value	Tolerance	Lead Free
RCT	01 : 0201		F (±1%)	
	02 : 0402		J (±5%)	
	03 : 0603			
	05 : 0805			
	06 : 1206			
	10 : 1210			
	12 : 1218			
	18 : 1812			
	20 : 2010			
	25 : 2512			

